INCH-POUND

MIL-B-49430/2E(ER)
1 August 1991
SUPERSEDING
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25 April 1988

MILITARY SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM SULFUR DIOXIDE BA-5598/U

This specification is approved for use within Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-B-49430(ER).

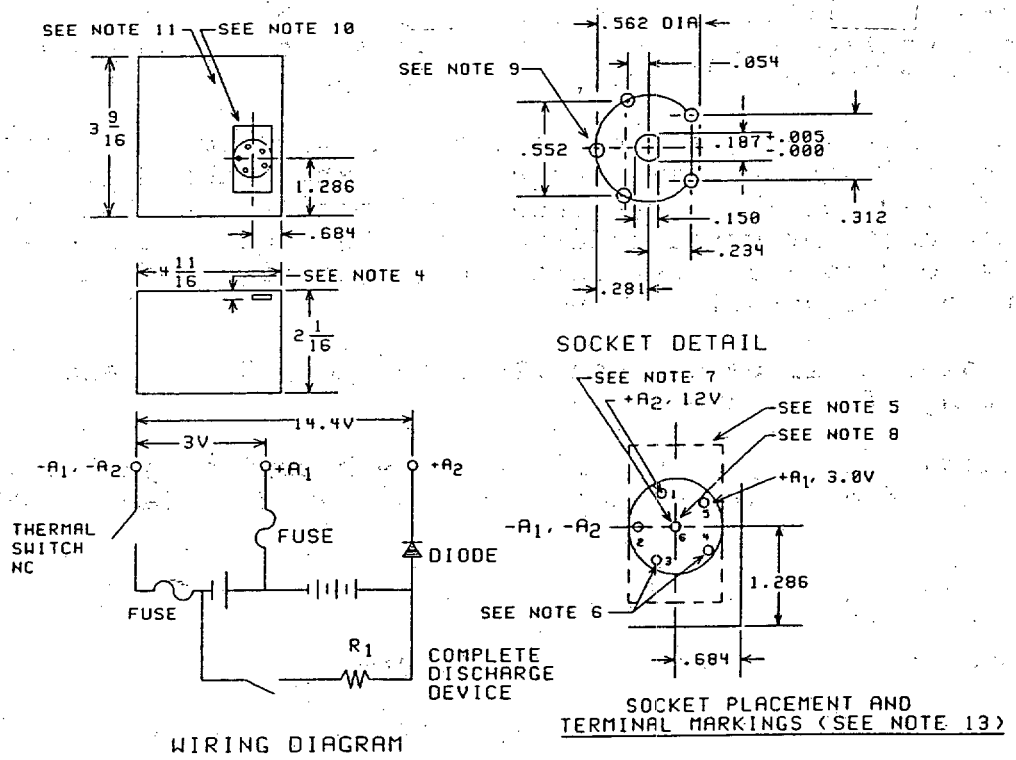


Figure 1. Battery dimensions and wiring schematic.

AMSC N/A FSC 6135

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NOTES:

1. All dimensions are in inches.

2. Tolerances, unless otherwise specified, are ±0.005 inch for decimals.

3. Tolerances for length, width, and height of battery are ±1/16 inch for fractions.

4. The minimum depth of socket well shall be 1/2 inch.

5. The size of the insulating plate is optional.

6. Socket insert contact not required.

7. Pin-circle center.

8. "D" hole for plug positioning.

9. Five holes to fit 1/8 inch pins, 7/16 inch in length.

10. Socket must float 3/32 + 1/32 -1/16 inch in all directions from the specified pin circle center location in the plane of the socket from reference dimension.

11. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during and after subsequent insertions of the mating plug during contractual testing.

12. R_1 value shall be between 125 ohms and 175 ohms and have a power rating of 2.0 watts.

13. Terminal marking to be printed or embossed on jacket in suitable location near the socket on the terminal face of the battery.

REQUIREMENTS:

Dimensions and configuration: See Figure 1.

	A_1 Unit	A_2 Unit
Maximum voltage:	3.05	15.25 volts
Minimum voltage:	2.00	10.00 volts

The battery shall contain a 2.25 ampere non-replaceable slow blow fuse in the negative leg and the $+A_1$ leg.

A diode shall be wired in series with the A_2 section. See paragraph 3.5.5.2 of the basic specification.

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Terminals: Five (5) hole socket type with no obstruction of any holes.

Weight (maximum): 1.5 pounds (681 grams)

Capacity tests: When the battery is tested in accordance with the methods of examination and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity Requirements in hours to 10.00 Volts
I	3.8
L	2.0
H	6.5
HT	6.3
LT	1.9
T	3.25
D	3.25

Initial voltage delay: When the battery is subjected to the I, H, HT, T, and D capacity tests, the time required at the beginning of discharge of the battery to reach a voltage of 10.00 volts after the specified load is applied shall not be more than 5.0 seconds. For the L and LT capacity tests the time to reach a voltage of 10.00 volts after the load is applied shall not be more than 60 seconds.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery in the negative leg as described in paragraphs 3.5.5 and 3.5.5.1 of the basic specification.

Complete discharge device: This device shall be inserted in the battery as shown on the circuit schematic and shall have the characteristics as described in paragraph 3.5.5.3 of the basic specification.

METHODS OF EXAMINATION AND TESTS:

Capacity tests:

(1) Storage: Details on storage conditions for all capacity tests are specified in basic specification (see 4.8.9).

(2) Discharge:

Cells: Discharge at 2.0 amperes constant current at 70±5°F for a minimum of 3.2 hours to 2.00 volts cut-off. Continue discharge to zero volts. Then discharge at 1.0 ampere constant current for 6.4 hours into voltage reversal.

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Batteries: The A_2 unit ($-A_1$, $-A_2$, to $+A_2$) shall be discharged as follows:
For all capacity tests, except the H and HT, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed-circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at one ampere constant current for 7.2 hours immediately after zero volts is obtained.

Closed-circuit voltage: Closed-circuit voltage measurements shall be made with a direct current voltmeter of proper range and sensitivity (see basic specification), using load resistance as shown in the following table:

UNIT RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE
A_1 6.5 ohms, $\pm$ 1%	1.9 volts
A_2 14.2 ohms, $\pm$ 1%	10.0 volts

Voltage shall be above "MINIMUM PERMISSIBLE VOLTAGE" within five (5) seconds after load is applied.

Socket strength test (First Article only): Before conducting the jacket integrity and visual mechanical examinations specified in Table III, the 5 batteries shall be subjected to the following:

Using a 0.250 $\pm$0.050 inch diameter rod, apply a force to the center of the socket and deflect it downward a distance of 0.250 +0.0625, -0 inch below the top surface of the battery.

Battery Label (only for this battery):

"DO NOT CHARGE, SHORT CIRCUIT, INCINERATE, MUTILATE OR TEST WITH U-410 ADAPTER OTHERWISE BATTERY MAY VENT OR RUPTURE RELEASING TOXIC MATERIALS"

Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodian:
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